JEREMY MATTATHIAS MBOE

AI Engineering Student

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SUMMARY

Artificial Intelligence Engineering undergraduate at Institut Teknologi Sepuluh Nopember with a strong foundation in machine learning, deep learning, and backend software development. Experienced in developing computer vision pipelines, temporal signal processing, and scalable REST APIs using Python, C++, and PyTorch. Proven technical leadership as a Teaching Assistant and Laboratory Assistant, driving rigorous model evaluation and applied research. Eager to leverage software engineering and intelligent systems to accelerate digital transformation, in-car innovation, and smart mobility solutions.

EDUCATION

Bachelor of Artificial Intelligence Engineering 08/2024 – Present | Surabaya, Indonesia
Institut Teknologi Sepuluh Nopember
- GPA: 3.86/4.00, Relevant Courses: Probability and Statistics, Computational Intelligence, Data Mining, Machine Learning, Deep Learning, Natural Language Processing, Digital Image Processing.

EXPERIENCE

Laboratory Assistant 05/2026 – Present | Surabaya, Indonesia
Intelligent Computing and Vision Laboratory
- Collaborate with laboratory researchers, faculty members, and clinicians from Rumah Sakit Universitas Airlangga on medical AI research, contributing to the development and evaluation of deep learning-based medical image segmentation pipelines for kidney cyst segmentation.
- Develop and organise practicum modules and instructional materials for AI-related courses, including Machine Learning, Data Mining, and related computational intelligence topics, supporting hands-on student learning.

Research Assistant 02/2026 – Present | Surabaya, Indonesia
Diagnostic and Prognostic Laboratory (D&P Lab) | AI Division
- Developed an automated TB Cough Diagnostic System utilizing LSTM architectures and acoustic feature extraction, achieving high-accuracy screening in clinical environments.

Teaching Assistant - Probability & Statistics, Machine Learning, Data Mining 02/2026 – Present | Surabaya, Indonesia
Department of Informatics Engineering, Institut Teknologi Sepuluh Nopember
- Facilitated tutorials and practicum sessions across Probability & Statistics, Machine Learning, and Data Mining, supporting students in both theoretical concepts and hands-on implementation.
- Guided students in statistical inference, data preprocessing, supervised and unsupervised learning, model evaluation, and exploratory data analysis.
- Developed instructional materials and programming exercises using Python, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, and PyTorch.
- Reviewed assignments and projects, providing technical feedback and troubleshooting support for data analysis and machine learning workflows.

Software Developer Intern 01/2026 – 02/2026 | Surabaya, Indonesia
Pusat Studi Kecerdasan Artificial dan Teknologi Digital Institut Teknologi Sepuluh Nopember
- Executed EEG signal preprocessing and multi-dimensional feature extraction, focusing on specific channels, frequency bands, and sub-bands for neurological data analysis.
- Developed and evaluated Machine Learning classifiers to interpret EEG patterns for diagnostic support.

PUBLICATIONS & PRESENTATIONS

e-Poster Presentation | ASEAN4TB Conference 01/07/2026
ASEAN4TB Conference
- Primary Author & Presenter, co-authored with ITS Department of Informatics Engineering & Department of Medical Technology.
- Developed an XGBoost model for early three-month mortality stratification in tuberculous meningitis, achieving an AUROC of 0.825.
- Built an interpretable eight-feature logistic regression surrogate using SHAP-selected predictors, achieving an AUROC of 0.815.
- Evaluated clinical utility with decision curve analysis, which indicated net benefit across threshold probabilities from 3 to 25 percent.

PROJECTS & RESEARCH

Multimodal TB Cough Detection & Diagnostic System 08/2025 – Present
Machine Learning Developer
- Developed an AI-driven Tuberculosis detection system utilizing LSTM integrated with acoustic signal processing.
- Awarded 1st Place & Best Scientific Paper at the National Electrolympic UPNVJ and presented technical methodologies to a panel of academic experts.

AVID-MM: Audio-Visual Identity Disentanglement 04/2026 – 07/2026
Lead Deep Learning Researcher
- Led development of a cross-modal speaker verification system designed to remain usable when audio or visual input is missing or corrupted.
- Built a shared embedding pipeline with ECAPA-TDNN voice features and pretrained face backbones, increasing accuracy from 59 percent to more than 93 percent on the POLY_1 challenge setup.

SwaraTTS: Latent Flow Matching (Avalon AI x RISTEK UI) 02/2026 – 06/2026
Deep Learning Researcher (Diffusion Models & Flow Matching Division)
- Contributed to a low-latency neural speech synthesis system using score-based diffusion and latent flow matching.
- Implemented and studied generative trajectories in latent space, including the relationship between diffusion processes and continuous normalizing flows.
- Evaluated sampling efficiency and generation quality without claiming state-of-the-art performance unless a verified benchmark supports that claim.

Foundational Neuromorphic Vision Research 04/2026 – 06/2026
Avalon AI Research Assistant

- Built a spiking neural network training pipeline in PyTorch from first principles, including custom surrogate gradients, temporal encoding, and spike-based layers.
- Implemented Izhikevich neuron dynamics and achieved 76.6 percent classification accuracy on N-MNIST without an external spiking framework.

EEG Signal Analysis and Binary Encoding 01/2026 – Present
Pusat Studi Kecerdasan Artifisial | AI Researcher
- Engineered an end-to-end signal processing pipeline to classify ALS vs. Normal subjects using 1,809 clinical EDF recordings.
- Developed a specialized feature extraction pipeline for segmented EEG tasks (Resting, Thinking, Typing), computing time-domain features (MAV, Variance, STD) across specific subbands (Delta through Gamma).

Oil Palm Tree Health Detection and Classification System 06/2025 – 08/2025
Deep Learning Researcher
- Collaborated with a team to develop a deep learning pipeline for detecting and classifying oil palm tree health from aerial imagery.
- Data Annotation & Processing: Led the large scale annotation effort for object detection, manually labeling hundreds of individual palm trees per high-resolution aerial image to ensure training set diversity and quality.
- Semantic Labeling: Developed a custom classification labeling workflow by applying NDVI (Normalized Difference Vegetation Index) masking to isolate tree canopies and accurately annotate health status based on spectral signatures.
- Implemented YOLOv8s for object detection and ResNet-18 for image classification with integrated NDVI masking to enhance vegetation features.
- Deployed the model on Hugging Face Spaces using Gradio, enabling interactive visualization and testing of results.

AWARDS

1st Place & Best Scientific Paper National Electrolympic UPNVJ (2025). 06/12/2025
Universitas Pembangunan Nasional Veteran Jakarta
Recognized for the technical research and paper presentation of the TB Cough Detection System.

SKILLS

Programming Languages: Python, C++, JavaScript, SQL

LLM and Generative AI Systems: LLM inference and serving, REST APIs, score-based diffusion, latent flow matching

Tools: Git, GitHub, Docker, Linux, Jupyter, PostgreSQL

Languages: English (Professional), Indonesian (Native).

AI/ML: PyTorch, TensorFlow, Hugging Face, scikit-learn, Pandas, NumPy, Matplotlib

CERTIFICATES

Dicoding : Machine Learning & Deep Learning Dev Certification for Machine Learning with TensorFlow

- Completed a hands-on program covering supervised and unsupervised learning, model training and evaluation, data preprocessing, and model deployment using TensorFlow.